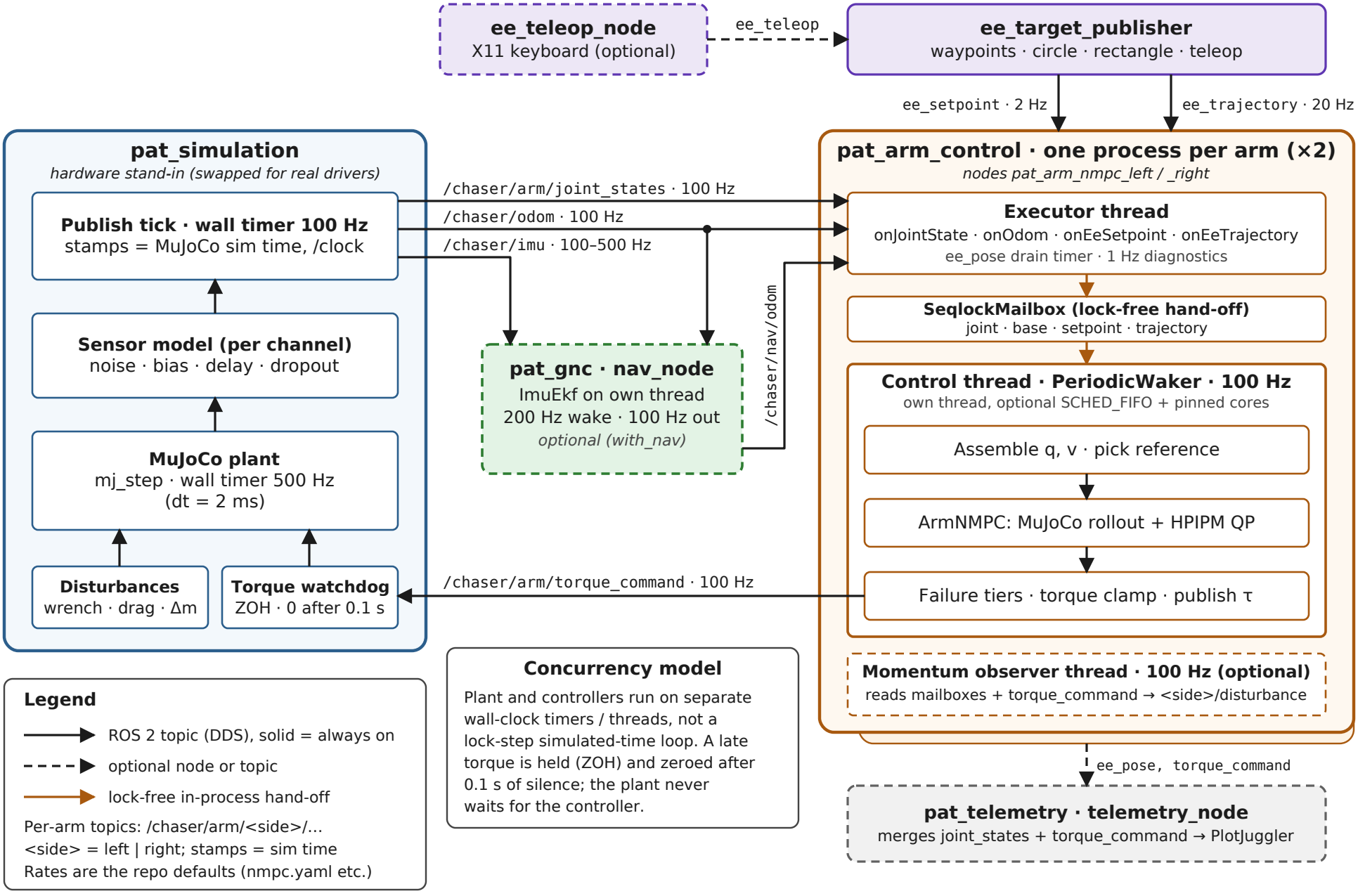
ee_teleop_node
X11 keyboard (optional)
ee_teleop
ee_target_publisher
waypoints · circle · rectangle · teleop
ee_setpoint · 2 Hz
ee_trajectory · 20 Hz
pat_simulation
hardware stand-in (swapped for real drivers)
Publish tick · wall timer 100 Hz
stamps = MuJoCo sim time, /clock
Sensor model (per channel)
noise · bias · delay · dropout
MuJoCo plant
mj_step · wall timer 500 Hz
(dt = 2 ms)
Disturbances
wrench · drag · Δm
Torque watchdog
ZOH · 0 after 0.1 s
/chaser/arm/joint_states · 100 Hz
/chaser/odom · 100 Hz
/chaser/imu · 100–500 Hz
pat_gnc · nav_node
ImuEkf on own thread
200 Hz wake · 100 Hz out
optional (with_nav)
/chaser/nav/odom
/chaser/arm/torque_command · 100 Hz
pat_arm_control · one process per arm (×2)
nodes pat_arm_nmpc_left / _right
Executor thread
onJointState · onOdom · onEeSetpoint · onEeTrajectory
ee_pose drain timer · 1 Hz diagnostics
SeqlockMailbox (lock-free hand-off)
joint · base · setpoint · trajectory
Control thread · PeriodicWaker · 100 Hz
own thread, optional SCHED_FIFO + pinned cores
Assemble q, v · pick reference
ArmNMPC: MuJoCo rollout + HPIPM QP
Failure tiers · torque clamp · publish τ
Momentum observer thread · 100 Hz (optional)
reads mailboxes + torque_command → <side>/disturbance
ee_pose, torque_command
pat_telemetry · telemetry_node
merges joint_states + torque_command → PlotJuggler
Legend
ROS 2 topic (DDS), solid = always on
optional node or topic
lock-free in-process hand-off
Per-arm topics: /chaser/arm/<side>/...
<side> = left | right; stamps = sim time
Rates are the repo defaults (nmpc.yaml etc.)
Concurrency model
Plant and controllers run on separate wall-clock timers / threads, not a lock-step simulated-time loop. A late torque is held (ZOH) and zeroed after 0.1 s of silence; the plant never waits for the controller.